Hunter Chase Velasquez

Design Engineer | Prototyping, CAD & Fabrication

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Purdue multidisciplinary engineer whose toy mechanisms are sold at Walmart and Target nationwide (Spin Master / Hatchimals), now designing and building vehicles and products for a 744K-subscriber YouTube channel—from CAD and laser-cut/CNC parts to welded assemblies tested in the field.

EXPERIENCE

Louis Weisz (YouTube, 744K subscribers) *Head of Production* — Boulder, CO | Fall 2025 – Present

Design engineering, CAD and fabrication, plus filming and editing · collaborator since 2022

- Design and build each project with Louis, handling most of the CAD and design-for-manufacture: jet-engine mounts for multiple vehicles (released and upcoming), a bent-steel e-bike hitch receiver, CNC aluminum control bar ends, and laser-cut vents, doors, and leveling legs.
- Prototype with FDM and SLA 3D printing, then design and order production parts through SendCutSend.
- Introduced a bill-of-materials and manufacturing-plan step **now standard on every build**; share fabrication work (welding, woodworking, assembly).
- Co-designed products sold as channel merch (Permaroid photo frame, flint striker); edit and film every release (3.4M+ views across the last 8).

Spin Master *Product Design Engineer* — Toronto (remote) | Summer 2024 – Summer 2025

- Designed and prototyped mechanisms for three Hatchimals Bloomables toys, **sold at Walmart and Target nationwide**.
- Brought onto the Kittyfly / Puppyfly interactive toy to simplify its petal-opening mechanism; redesigned it from **50+ parts and features to under 20**.
- Designed the mechanism for the Bloom Bash Terrarium's water-activated pop-up reveal and took it through design reviews to production; prototyped concepts for the Flower Fest stage playset.
- Worked with project managers, graphic designers, and engineers through rounds of documented prototypes.

The RSVP Group *Video Media Manager* — Brunswick, OH | 2020 – 2024

- Ran video production and publishing: videography, photography, and editing.

Strategic Productions LLC *Media Production Engineer* — Boulder, CO | Summer 2023

- Led design and builds for a 700K+ subscriber channel, e.g. electric vs. gas ride-on Jeeps: welded chassis, 3000W electric and 212cc gas drivetrains, rack-and-pinion steering.

Asterbrook Properties Rehab *Home Remodeling Apprentice* — Brunswick, OH | 2017 – 2022

- Renovated distressed homes: structural, electrical, and plumbing work alongside owners and contractors.

SELECTED BUILDS

- **Jet-Powered Longboard (for Sickos)** — Co-designed a turbine-powered skateboard for outside creators: wheel-clearing jet mount (3D-printed prototypes, then laser-cut parts), baffled backpack fuel tank, truck geometry tuned against speed wobble.
- **E-Bike Micro Camper (2.2M views)** — Telescoping camper that doubles in length on drawer slides; custom bent-steel hitch receiver for a bike no one makes a hitch for; later added adjustable leveling legs that doubled as a jack for a trail repair.
- **Stealth Electrical Box (588K views)** — Lightweight fake utility box with a hidden sliding door, epoxy-skinned foam panels that pass as sheet metal, and resin-printed hinges; went undetected for 24 hours in a parking lot.
- **JAWS** — Lead engineer on a pneumatic rig that reproduces carnivore bite marks on bone; sand-cast aluminum jaw. Presented at the SAA 87th Annual Meeting; 2nd and 3rd place, Purdue Undergraduate Research Conference (2022).

EDUCATION

Purdue University B.S.E. Multidisciplinary Engineering, Minor in Theatre Design & Production — West Lafayette, IN | 2024

SKILLS

CAD: SolidWorks, Fusion 360, Creo, NX, AutoCAD, Vectorworks
Prototyping & manufacturing: FDM and SLA 3D printing, laser-cut / CNC part design, BOMs and build plans
Fabrication: Welding, metalworking, woodworking, sand casting, CNC machining, home remodeling
Other: Python, MATLAB, R; Adobe Premiere Pro, After Effects, Photoshop, Lightroom & Creative Cloud